

Geometry from the Inside Out

*The Dot Product as Area, Dimension as Intersection,
and Zero as the Horizon of Geometry*

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*"The question, asked patiently and followed wherever it leads,
is what mathematics actually is."*

Abstract

This article develops a series of interconnected geometric and philosophical insights originating from a single visual construction: a triangle formed by three vectors summing to zero, with its squared side lengths partitioned into rectangles whose signed areas are the negatives of the pairwise dot products of the edge vectors.

From this construction flows a cascade of ideas: the dot product as a geometric area intrinsic to the triangle rather than an externally imposed algebraic operation; the Law of Cosines as a trivial consequence of the distributive property of the inner product; a novel triangle invariant — the **Pourkavoos Invariant** — not previously named in the standard literature; the inversion of the standard dimensional hierarchy; the reframing of intersection rather than construction as the generative operation of geometry; the structural equivalence of zero and infinity as the twin unreachable horizons bounding a dimensional continuum; and deep connections to projective geometry, exterior algebra, geometric algebra, Minkowski spacetime, quantum mechanics, general relativity, and category theory.

None of these ideas require advanced mathematics to appreciate. All of them emerge from looking carefully at three vectors and three squares.

Contents

1 The Triangle, the Squares, and the Rectangles

1.1 The Construction

Begin with a triangle — not an abstract one, but a specific triangle built from three vectors. Label the edge vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$. These three vectors are not arbitrary: they satisfy

$$\vec{a} + \vec{b} + \vec{c} = \vec{0}. \quad (1)$$

Equation (1) means the vectors form a closed loop — walking along $\vec{a}$, then $\vec{b}$, then $\vec{c}$ returns you exactly to the starting point. Every triangle in the plane can be expressed this way.

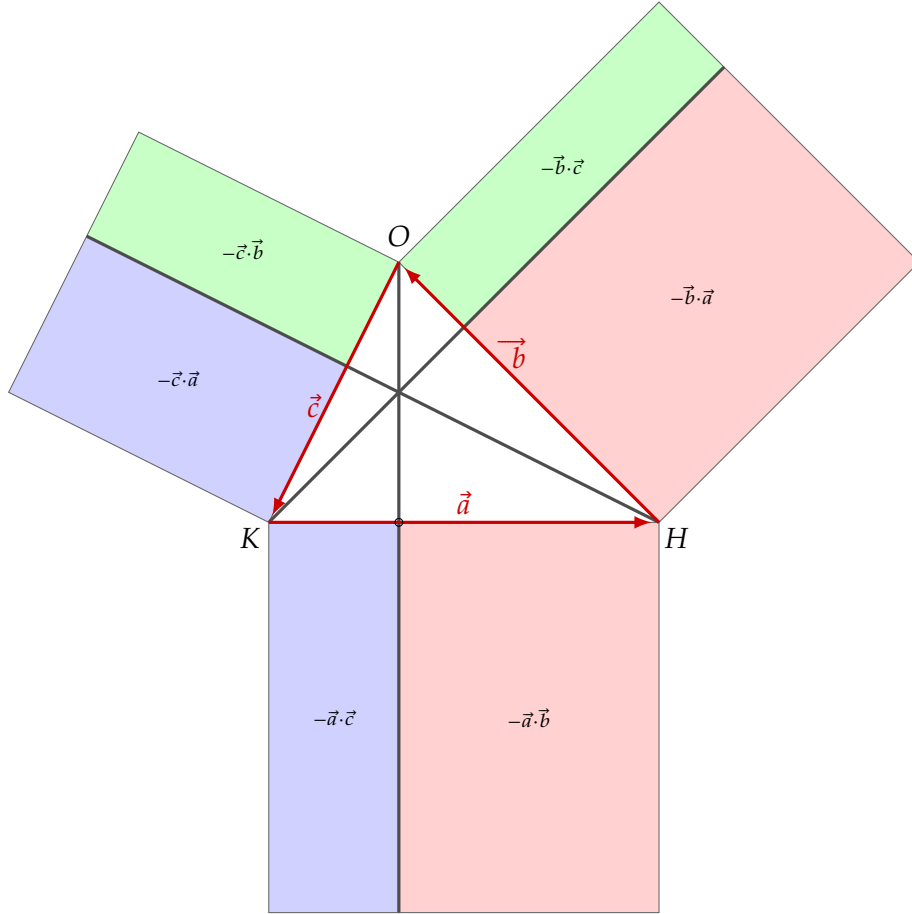


Figure 1: The central construction. Triangle KHO with edge vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ satisfying $\vec{a} + \vec{b} + \vec{c} = \vec{0}$. Each square on a side is partitioned into two coloured rectangles whose signed areas are the negatives of the pairwise dot products. The three altitudes pass through a common point (the orthocentre).

On each side of the triangle, construct a square. The square on side $\vec{a}$ has area $|\vec{a}|^2 =$

$\vec{a} \cdot \vec{a}$, and likewise for $\vec{b}$ and $\vec{c}$. Now ask: can each square be partitioned into pieces whose areas involve the *other* sides?

Because $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, dotting both sides with $\vec{a}$ gives

$$\vec{a} \cdot (\vec{a} + \vec{b} + \vec{c}) = 0,$$

which expands to

$$\vec{a} \cdot \vec{a} + \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c} = 0.$$

Rearranging:

$$\vec{a} \cdot \vec{a} = (-\vec{a} \cdot \vec{b}) + (-\vec{a} \cdot \vec{c}). \quad (2)$$

Applying the same argument cyclically:

$$\vec{b} \cdot \vec{b} = (-\vec{b} \cdot \vec{c}) + (-\vec{b} \cdot \vec{a}), \quad (3)$$

$$\vec{c} \cdot \vec{c} = (-\vec{c} \cdot \vec{a}) + (-\vec{c} \cdot \vec{b}). \quad (4)$$

Each squared side length equals the sum of exactly two *negated* dot products of the other edge vectors. These six quantities are the areas of the six coloured rectangles visible in Figure ?? . They tile the three squares completely and without overlap.

1.2 Why the Rectangles Are Negative

A careful reader will notice that the coloured rectangles in Figure ?? have positive geometric area, yet the algebra labels them $-\vec{a} \cdot \vec{b}$, $-\vec{b} \cdot \vec{c}$, and so on — that is, negatives of dot products. This means the dot products themselves are *negative*. Why?

The angle between $\vec{a}$ and $\vec{b}$, measured tail to tail, is $\pi - \alpha$ where α is the interior angle at H , giving

$$\vec{a} \cdot \vec{b} = -|\vec{a}||\vec{b}| \cos \alpha.$$

For an acute triangle all interior angles satisfy $\cos \alpha > 0$, so all three dot products are negative and all six rectangle areas $-\vec{a} \cdot \vec{b}$ are positive. For an obtuse triangle the one dot product corresponding to the obtuse angle becomes positive, and its negation is still the correct positive rectangle area. The construction holds in all cases.

1.3 The Law of Cosines in Two Lines

The standard proof of the Law of Cosines drops an altitude, invokes the Pythagorean theorem twice, and assembles the result algebraically. The present construction renders all of that unnecessary.

Denote $a^2 = \vec{a} \cdot \vec{a}$, $b^2 = \vec{b} \cdot \vec{b}$, $c^2 = \vec{c} \cdot \vec{c}$. Substituting equations (??) and (??) into (??), and using the symmetry $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$:

$$c^2 = a^2 + b^2 - 2(-\vec{a} \cdot \vec{b}).$$

Since $-\vec{a} \cdot \vec{b} = ab \cos C$ (where C is the interior angle at vertex H between sides a and b), this is exactly

$$\boxed{c^2 = a^2 + b^2 - 2ab \cos C.} \quad (5)$$

This is the Law of Cosines. It has not been derived from trigonometry; it has been *revealed* as a direct consequence of the distributive property of the inner product applied to the closure condition $\vec{a} + \vec{b} + \vec{c} = \vec{0}$. The trigonometry is secondary; the dot product identity is primary.

The Pythagorean theorem emerges as the special case $\vec{a} \cdot \vec{b} = 0$ (right angle at H), in which one rectangle collapses to zero width and the three-square identity reduces to $c^2 = a^2 + b^2$.

1.4 The Pourkavoos Invariant

The construction yields a further result. Consider the three quantities

$$a^2 - \vec{b} \cdot \vec{c}, \quad b^2 - \vec{c} \cdot \vec{a}, \quad c^2 - \vec{a} \cdot \vec{b}.$$

From equations (??)–(??) one can verify that all three are equal, and that their common value equals the symmetric combination $-(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$.

The Pourkavoos Invariant. For any triangle with edge vectors $\vec{a}, \vec{b}, \vec{c}$ satisfying $\vec{a} + \vec{b} + \vec{c} = \vec{0}$:

$$-(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = a^2 - \vec{b} \cdot \vec{c} = b^2 - \vec{c} \cdot \vec{a} = c^2 - \vec{a} \cdot \vec{b}. \quad (6)$$

In any triangle, the area of the square on a side, plus the signed area of one of its corresponding dot-product rectangles, is the same for all three sides.

In plain language: the Pourkavoos Invariant is **a single number that characterises the triangle as a whole**, not any individual side. Every side reveals it through a different decomposition; all three decompositions agree.

In the special case of a right triangle ($\vec{a} \cdot \vec{b} = 0$ at vertex H), the invariant reduces to c^2 — the square of the hypotenuse — confirming that the Pythagorean theorem is

the right-angle instance of the more general identity.

This invariant does not appear under a standard name in the classical triangle geometry literature. It may be related to known triangle quantities such as the circumradius, the area, or the sum of squared sides, and that connection deserves further investigation.

2 The Dot Product Is an Area

2.1 Two Ways of Seeing the Dot Product

The standard introduction to the dot product presents it in one of two ways: either algebraically (multiply components, sum) or trigonometrically ($|\vec{a}||\vec{b}|\cos\theta$). In both cases the result is a scalar — a single dimensionless number.

The construction of Section 1 suggests a different reading. Each term $-\vec{a} \cdot \vec{b}$ is not merely a number. It is the area of a specific rectangle inscribed in the geometry of the triangle. The dot product has a geometric *home* — it does not float free as an abstract scalar but lives as a visible, coloured region inside one of the three squares.

This shift may seem cosmetic, but it is not. In the standard view, you apply an algebraic operation to geometric objects to produce a number. In the view offered by Figure ??, the number is already present in the geometry, waiting to be identified. The dot product is the act of finding and measuring that rectangle.

2.2 Synthesis and Analysis: Two Directions Through Dimension

There are two natural directions of travel through the dimensional hierarchy, and they are not equivalent.

Synthesis (construction, building upward): points combine to form lines; lines combine to form planes; planes sweep out volumes. Each level is assembled from the level below. Dimension *increases* as you build.

Analysis (projection, extracting information): a volume is characterised by its cross-sectional areas; a plane by distances to a fixed line; a line by scalar measurements along it. Dimension *decreases* as you extract.

The dot product belongs firmly to the second direction. It takes two vectors (one-dimensional objects) and extracts from their relationship a scalar — the zero-dimensional

measure of the oriented area they span. It is an analytical operation, not a synthetic one.

Standard pedagogy almost universally presents geometry in the synthetic direction: points are primitive, and everything is built from them. The diagram in Figure ??, by making the areas visible before the scalars, suggests that the analytic direction is equally natural — and perhaps more honest about what the dot product *is*.

2.3 The Area as the more Primitive Object

Push this idea further. **What if the rectangle is the more primitive object compared to lines?**

In arithmetic, the number 12 can be written as 3×4 or 6×2 . The number is primary; the factors are secondary. But one could equally imagine a geometry where the rectangle of area 12 is primary, and the dimensions 3 and 4 are the two ways of reading its structure along different axes.

In the triangle construction, **the rectangle** $-\vec{a} \cdot \vec{b}$ is not *produced* by the vectors $\vec{a}$ and $\vec{b}$. **It is the relationship** between them, expressed as geometric extent. The vectors are the factors. The area is the relationship itself.

This resonates immediately with quantum mechanics, where what is physically real is not the state of a particle in isolation but the relationship between its state and a measurement context — an inner product. Reality, in quantum theory, is relational. The dot product is the measure of that relation.

3 Inverting the Dimensional Hierarchy

3.1 The Standard Hierarchy and Its Hidden Assumption

The conventional dimensional hierarchy, inherited from Euclid, reads:

$$\text{Point (0D)} \rightarrow \text{Line (1D)} \rightarrow \text{Plane (2D)} \rightarrow \text{Volume (3D)} \rightarrow \dots$$

Each object is built from lower-dimensional components. A line is swept by a moving point. A plane is swept by a moving line. The point is the atom — the indivisible primitive from which everything is constructed.

This hierarchy feels natural because it mirrors how humans think about construction:

start simple, combine into complex. But it embeds a **hidden assumption: that lower-dimensional objects are more fundamental than higher-dimensional ones**. The dot product construction invites us to question that assumption.

3.2 Intersection as the Generative Operation

Consider the following **alternative definitions, running in the opposite direction**:

- A **point** is the *intersection* of two lines. It exists because two lines meet. Remove either line, and the point has no definition. It is purely the mark of their encounter.
- A **line** is the *intersection* of two planes. The crease where two flat surfaces meet is a line. It exists because the planes agree and diverge simultaneously at that one-dimensional locus.
- A **plane** is the *intersection* of two volumes — the septum between two soap bubbles. It comes into being only because two volumes press against each other and negotiate a shared boundary.
- A **volume** is the *intersection* of two four-dimensional regions, manifested as a three-dimensional boundary.

In this framework the hierarchy reads:

$$\dots \rightarrow \text{Volume (3D)} \rightarrow \text{Plane (2D)} \rightarrow \text{Line (1D)} \rightarrow \text{Point (0D)}$$

Lower-dimensional objects are *more derived*, not more primitive. They are the signatures of encounters between higher-dimensional things.

3.3 The Soap Bubble as Philosophical Demonstration

The soap bubble image is not merely an analogy — it is a physical realisation of intersection geometry.

When two soap bubbles touch, they form a shared wall, the septum, which is very nearly a flat plane. This plane did not exist before the bubbles touched. It came into being as the physical expression of the interface between two volumes, ontologically dependent on both. It exists precisely because two volumes are in contact and in tension. It is their *intersection*.

The plane, furthermore, knows where it is because the volumes tell it, by meeting there. The same is true for lines and points in the intersection hierarchy. A line

knows where it is because two planes tell it. A point knows where it is because two lines agree on it.

This is the key philosophical principle: **existence is a consequence of encounter, not a precondition for it.**

3.4 Connection to Projective Geometry

This idea has deep roots in the history of mathematics. Projective geometry — developed by Desargues and Poncelet and brought to maturity in the 19th century — adopts exactly this perspective. In the projective plane, a point is not a primitive: it is defined as the intersection of two lines.

4 Zero and Infinity as the Twin Horizons of Geometry

4.1 The Question the Hierarchy Cannot Answer

The intersection hierarchy **descends from high dimensions toward low**. Volumes are the intersections of four-dimensional regions. Planes are the intersections of volumes. Lines are the intersections of planes. Points are the intersections of lines.

4.2 Zero Functions Like Infinity

Now compare this to the behavior at the other end of the number line. What is $\infty + \infty$? The answer is still ∞ — not a larger infinity (in the ordinary arithmetic sense) but the same boundlessness repeated. The operation stops producing new structure. You are already at the limit; adding more of the limit returns the limit.

The structural parallel is exact. **Zero, at the bottom of the dimensional hierarchy, behaves exactly as infinity behaves at the top of the number line.** Both are places where the normal generative operation — intersection going down, extension going up — ceases to produce new structure. Both are limits that can be approached but not passed through.

Zero is not the beginning of geometry. Zero is the lower horizon of geometry, just as infinity is the upper horizon.

4.3 Rethinking What Zero Is

In standard mathematics, zero is often treated as the simplest thing: the empty set, the additive identity, the origin, the nothing-from-which- you-build. Dimension zero means a point — the most primitive geometric object. Zero is the ground floor.

In the intersection hierarchy, every object is defined by the objects “considered” above it: a point exists because two lines meet there, not prior to them. Points are therefore the most *derived* objects in the hierarchy — the result of the most intersections — not the most primitive. They are not where geometry starts; they are where the descent through dimensions reaches its limit. The operation does not break down *after* points — it breaks down *at* them, when you try to go one step further.

Zero is not where geometry starts. Zero is where geometry runs out of structure.

This parallels the behaviour of **infinity in the number system**. Infinity is not where counting begins — it is where counting ends. You can count to any finite number, but infinity remains always ahead, unreachable. Similarly, you can intersect to any positive dimension, but you cannot descend below zero; the operation fails before you arrive.

In this sense, **a point is to geometry what infinity is to algebra**. It is not a starting primitive but an unreachable limit, approached but never truly arrived at.

Yes, **A point is to geometry what infinity is to algebra!**

Would this then mean, that zero in algebra is somehow isomorphic to infinite dimensions in geometry?

4.4 The Riemann Sphere: Formal Confirmation

This philosophical intuition finds a precise mathematical expression in the theory of the Riemann sphere. The Riemann sphere is a way of extending the complex number plane to include a single point at infinity, creating a compact space in which zero and infinity play completely symmetric roles.

The key transformation is $z \mapsto 1/z$, which sends every complex number to its reciprocal. Under this map, $0 \mapsto \infty$ and $\infty \mapsto 0$. The unit circle is fixed. Every other point is reflected across it, with small numbers mapping to large ones and vice versa.

The function $1/z$ treats zero and infinity as *mirrors* of each other. The Riemann sphere makes this precise by placing them at the two poles of a sphere, with the rest of the complex plane wrapped around the equator between them. They are not different in kind. They are related by a simple inversion — structurally identical, positionally

opposite.

The insight of this article arrives at the same structural conclusion by a completely different path: Why is the dot product of two one-dimensional objects a zero-dimensional scalar? The dot product takes two vectors — one-dimensional objects — and produces a scalar, a zero-dimensional object. Each factor contributes a descent of one dimension; two factors bring us all the way to zero. But zero, as we have argued, is not a neutral ground floor — it is **the lowest horizon** of geometric intersection hierarchy.

All natural mathematics, in this view, lives between zero and infinity as the twin outer bounds of what it is real, geometrically and algebraically.

5 Connections, Echoes, and Extensions

5.1 Category Theory and the Primacy of Relations

The philosophical idea running through the **intersection hierarchy** — that objects are defined by their relations, not prior to them — finds its most precise mathematical expression in category theory.

Category theory, developed by Eilenberg and Mac Lane in the 1940s, organises mathematical structures not by their internal composition but by the maps (morphisms) between them. An object is characterised entirely by its relationships to other objects. The canonical example: the internal structure of a set can be reconstructed entirely from the pattern of functions from that set to other sets. Elements are not primitive; they are inferred from the relational pattern.

This is the exact thesis of the intersection hierarchy. A point is not a primitive atom; it is an object characterised by its relationships to two lines. A line is characterised by its relationships to two planes. Remove the relationships, and the objects have no definition.

Category theory and the **intersection hierarchy** are both expressions of the same deep principle: **in mathematics, as in physics, being is relational.**

6 Synthesis and Open Questions

6.1 What the Diagram Teaches

It is worth stepping back to take stock of how much has followed from a single diagram — a triangle, three squares, and six coloured rectangles.

The dot product is not merely an algebraic operation. It is a geometric object: a signed area inscribed in the structure of the triangle. The Law of Cosines is not a theorem about trigonometry; it is a theorem about inner products, of which trigonometry is a special case.

The Pourkavoos Invariant, the equal value of $-(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$ across all three sides, is a symmetric triangle invariant not previously named in the standard literature.

The dot product construction suggests an alternative dimensional hierarchy in which areas, not points, are natural intermediary objects. The intersection hierarchy defines objects relationally rather than constructively, connecting to projective geometry, category theory, and the philosophy of relational existence. And zero and infinity are structural twins. They are the two unreachable horizons bounding the dimensional continuum.

6.2 Open Questions

Several threads raised in this article invite further investigation.

The Pourkavoos Invariant $-(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$ deserves a classical geometric interpretation. Does it equal four times the triangle's area? Is it a function of the circumradius or inradius? The answer may exist somewhere in the classical literature on triangle geometry, but it does not appear in standard textbooks under this form. Connecting it to known invariants would complete the picture.

The intersection hierarchy, generalised to higher dimensions, suggests a definition of dimension purely in terms of the intersection operation, without reference to coordinates or metric structure. Can this be made rigorous? Is it equivalent to the standard topological or linear-algebraic definitions of dimension in appropriate contexts?

What is the right formalism for capturing the idea that the dimensional continuum is bounded at both ends — compact with respect to the intersection and construction operations?

And perhaps the deepest open question: if existence is relational, what generates the highest-dimensional objects from which all lower-dimensional ones descend?

6.3 A Final Reflection

Mathematics has a way of hiding its most beautiful ideas in plain sight. The dot product is introduced in every linear algebra course as a computational tool: multiply components, sum, done. The triangle is introduced in every geometry course as a figure with three sides and three angles. The fact that these two objects are deeply, visually, structurally intertwined — that the dot product lives as coloured rectangles inside the very squares that the triangle generates — is not a secret. But it is not shown either.

Figure ?? shows it. And from that showing, a cascade of ideas follows: about the nature of dimension, about the role of intersection in defining geometric objects, about the structural identity of zero and infinity, about the relational character of mathematical existence, about the unity of geometry and algebra.

None of these ideas required a computer, or a long calculation, or advanced machinery. They required looking carefully at a triangle and asking: *what is really going on here?* That question, asked patiently and followed wherever it leads, is what mathematics actually is.

I have been thinking about the geometric meaning of the dot product and its implications for the foundations of geometry for a long time. This article represents the synthesis of that exploration, beginning from a single diagram of three vectors, and following the ideas wherever they lead. The original diagram was typeset in L^AT_EX , and completed by me on February 23 , 2026.